

Long Nguyen Van

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Summary

Senior Data Engineer with 7+ years of experience in SQL, database design and ETL/data warehousing, building batch and real-time (CDC) pipelines with Spark, Kafka/Debezium and Airflow across Oracle, MySQL and PostgreSQL, plus the internal applications and APIs that serve data to business teams. Currently at a securities company, delivering reliable reporting data for trading, finance and risk users, and using Generative AI both in the development workflow and in products (MCP server, text-to-SQL "Virtual DA").

Experience

Techcom Securities JSC, Senior Data Engineer

Ha Noi, Vietnam
Oct 2022 – Present

- Built and operate real-time CDC pipelines with Kafka, Debezium and Kafka Connect, replicating Oracle, MySQL and PostgreSQL data into the reporting layer so departments get up-to-date data instead of next-day batch loads
- Develop and tune Apache Spark batch and streaming jobs that aggregate data from databases, APIs, Elasticsearch and MongoDB into report-ready datasets
- Designed cloud pipelines on AWS Glue, Lambda and Redshift, and collected data from unstructured sources (websites, documents, APIs)
- Build internal applications and REST APIs that give other departments self-service, access-controlled data, replacing manual data requests
- Share data securely with banking partner departments, with Transparent Data Encryption (TDE) and role-based access control (RBAC) applied at the database level
- Established data governance: data dictionary, business glossary, governance tooling and regular data quality reports to track improvements
- Partner with analysts and data scientists to turn business requirements into datasets and reports
- Use AI tools (Claude, Kiro, Gemini) daily across the development lifecycle for coding, SQL and pipeline development, debugging and documentation, and always review and validate AI output before use
- Built an MCP (Model Context Protocol) server exposing the data dictionary and business glossary, so other departments and AI applications can discover data definitions programmatically
- Built "Virtual DA", an internal text-to-SQL application on top of the MCP data dictionary: users ask questions in natural language, the system generates and executes the query on the database and returns the result, reducing ad-hoc requests to the data team

Tech Stack: Spark, Kafka, Debezium, Airflow, Oracle, PostgreSQL, AWS Glue/Lambda/Redshift, Python, Java, Jenkins, MCP, LLM

OpenCommerce Group, Data Engineer

Ha Noi, Vietnam
Dec 2021 – Oct 2022

- Built the WarehouseGen2 data warehouse on Amazon Redshift in a 3-person team, with multi-source ingestion via API, crawlers, webhooks, Kinesis Firehose and AWS DMS (PostgreSQL, MySQL to Redshift)
- Self-hosted and managed Airflow for orchestration, and implemented data models and data quality tests with dbt
- Migrated legacy databases to Redshift, designed data models and data marts, and tuned slow queries and database workloads
- Supported reporting on Holistics, Power BI and Tableau

Tech Stack: Python, Airflow, dbt, Redshift, PostgreSQL, AWS, Docker, Kinesis Firehose

Giaohangtietkiem JSC, Data Engineer

Ha Noi, Vietnam
Aug 2019 – Nov 2021

- Built Kafka connectors capturing database binlog events, and Spark Streaming ingestion from Kafka to HDFS
- Created ETL pipelines aggregating data from HDFS into MySQL and MongoDB, and resolved Kafka and Spark streaming performance issues
- Built **Data as Services** (2-person team): a query-builder app letting non-technical users create reports, with RBAC, and scheduled XLSX/PDF delivery by email and chat; reduced 80% of requests to the Data Analyst team, serving hundreds of users
- Deployed Superset with OAuth authentication, and built AI models for product classification and user trend analysis

Tech Stack: Java, Scala, Python, Hadoop, Spark, Kafka, Flask, Spring Boot, ReactJS, MySQL, ElasticSearch

NERD JSC, NLP Researcher Intern

Ha Noi, Vietnam
Aug 2018 – Aug 2019

- Contributed to **KAMI**, a Vietnamese-language virtual assistant launched in July 2019 for exam season (score lookup, admission search, school recommendations), [featured by VNU](#) 
- Improved intent detection for the assistant, built crawlers to collect training and admission data, and worked on Named Entity Recognition and diacritic restoration

Tech Stack: Python, NLP, Named Entity Recognition, intent detection, web crawling

Skills

- Databases: Oracle, MySQL, PostgreSQL, Amazon Redshift, MongoDB, ElasticSearch; database design, data modeling, query tuning, data migration
- Languages: SQL, Python, Java, Scala, JavaScript
- ETL & CDC: Kafka, Kafka Connect, Debezium, Apache Spark (batch, Structured Streaming), Airflow, dbt, AWS Glue, AWS DMS
- Cloud: AWS (Glue, Lambda, Redshift, Kinesis, DMS), Docker
- Serving: Spring Boot, Flask, REST APIs, ReactJS; BI: Superset, Redash, Power BI, Tableau, Holistics
- AI: LLM applications, MCP servers, text-to-SQL, prompt engineering, NLP
- DevOps & AI tools: Jenkins (CI/CD), Git, Claude, Kiro, Gemini; code review

Education

VNU University of Engineering and Technology, BSc Information Systems

2016 – 2020

- Thesis: Automatic Diacritic Restoration of Vietnamese Text (99% accuracy)

Languages

English: Professional working proficiency | Japanese: JLPT N4 (2018)